

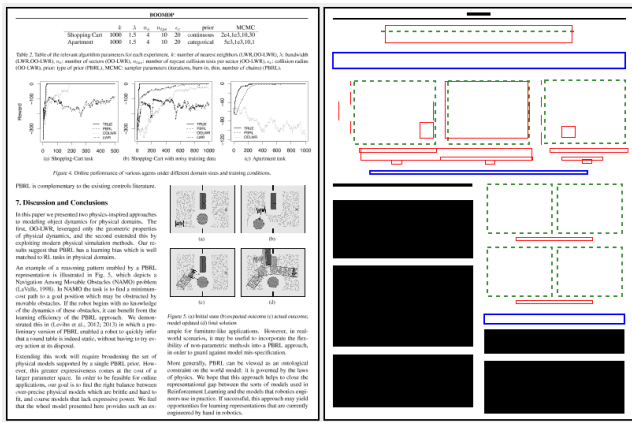


Figure 1: A document page (left panel, from [Scholz et al., ICML 2014]) decomposed into a set of classified regions (right panel). Body text regions are shown as filled boxes, captions and figure text regions as box outlines, and graphical element regions as dashed box outlines.

Region Identification.

Region identification decomposes document pages into regions, each one labelled as either caption, graphical element, body text, or figure text. Caption regions are built by starting from the caption phrases found in the prior step, and combining them with subsequent lines of text. The rest of the text in the document is grouped into paragraphs using Poppler’s paragraph grouping mechanism. Paragraphs that are either too large or aligned to the left margin of a column are classified as body text, otherwise they are classified as figure text.

Page headers and page numbers are handled as special cases. PDFFigures checks if pages in the document are consistently headed by the same phrase, and if so marks those phrases as body text. Likewise page numbers are detected by checking if all pages end with a number, and if so marking those numbers as body text.

Finally, the graphical elements of the document are located. To do this each page is rendered as a 2D image using a customized PDF renderer that ignores text. The bounding boxes of the connected components in the resulting image are then used as graphical regions of the document. An example of such a decomposition is shown in Figure 1.

Figure Assignment.

The last step is to assign each caption a region of the document containing the figure it refers to. First, up to four “proposal” regions are generated for each caption. Proposal regions are built by generating a rectangular region adjacent to each side of the caption, and then maximally expanding those regions as long as they do not overflow the page margin, overlap with body text, or overlap a caption. This is shown in Figure 2. For two-column papers regions are constrained to not cross the center of the page unless the caption itself spans both columns.

Next, a single proposed figure region is selected for each caption. To do this a scoring function is used to rate each proposed region based on how likely it is to contain a figure.

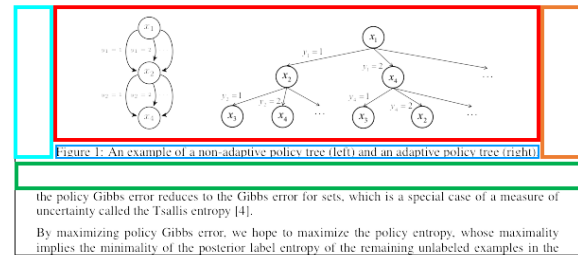


Figure 2: Generating possible figure regions for a caption. For each caption (center blue box), up to four regions are generated as possible figures by maximally expanding boxes that are adjacent to one side of the caption. Page from [Cuong et al., NIPS 2013]

The scoring function gives higher scores to regions that are large and contain graphical elements. To ensure captions are not assigned regions that overlap, they iterate through every possible permutation of how figure regions could be matched to captions. Each permutation is scored based on the sum of the scores of the proposed regions it includes, with regions that overlap given a score of 0. The highest scoring permutation is then selected as the final set of figure regions to return.

An additional complication comes from cases where figures are immediately adjacent, so that they are not separated by any intervening body text or captions. In these cases, proposal regions might get overly expanded and therefore contain multiple figures. To handle these cases, when iterating through permutations, if two proposal regions overlap an attempt is made to split them by detecting a central band of whitespace that separates them. An example of such a figure can be found in Figure 4, second row right column.

4. PROPOSED APPROACH

Our approach builds upon [5] by making crucial updates to its important components. Most of the updates are designed to allow PDFFigures 2.0 to generalize across a wider variety of paper formats. The PDFBox [1] library is used for parsing PDFs.

4.1 Caption Detection

We improve the keyword search of [5] that identifies phrases that might start captions to be effective against more kinds of papers by using a considerably expanded set of keywords. However, naively increasing the number of keywords also increases the number of false positives for each paper. We resolve this problem by adding a number of additional filters to the ones used in [5], such as (I): Select phrases that are all caps. (II): Select phrases that are abbreviated. (III): Select phrases that occupy a single line. (IV): Select phrases that do not use the most commonly used font in the document. (V): Select phrases that are left aligned to the text beneath them. The last filter serves as a general purpose filter for detecting indented paragraphs or bullet points that start by mentioning a figure.

4.2 Region Identification

Our region identification method decomposes each page into caption, body text, figure text and graphical regions as